

		Hence $\frac{m_X}{m_Y} = \frac{V_X}{V_Y} \frac{T_Y}{T_X} = 2 \left(\frac{400}{200} \right) = 4$ so, $m_Y = \frac{m}{4}$
11	B	work done = $p\Delta V = p_f V_f - p_i V_i = nRT_f - nRT_i$ Since the process happens at constant pressure when volume doubles, temperature doubles. so work done = $p\Delta V = nRT_f - nRT_i = nR(2T - T) = nRT$ work done = RT when $n = 1$
12	C	Option A, B and D are not correct. Internal energy is also affected by the